

Extrema-Frame Foreground-Background Theory II: Realization Universality, Global Complexity, and Correlated Layered Semantics

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September 2026

Abstract

Extrema-Frame Foreground-Background theory (EF-FG) separates observer-relative scalar-field structure into a spatial-frame projection and a polarity-core ownership projection, and evaluates uncertainty through exact common-realization semantics. This sequel resolves the theorem-sized directions identified in the foundational paper. First, we classify the global spatial-frame realization problem: a native terminal-mixed-mask query is NP-complete on bipartite radius-two graphs, and realization of a fully prescribed finite categorical spatial-frame state is NP-complete on bipartite graphs of radius at most three. Second, we determine the realization class of protected-obligation complexes exactly: under both raw and canonical semantics, every finite simplicial complex occurs as an EF-FG common-realization complex. Third, we identify two fixed-parameter tractable escape routes from the ownership hardness kernel. If k is the number of elder-order-ambiguous owner events, q-band completion is decidable in $O(k! \text{poly}(n))$ time; if g is the number of nondegenerate attachment/gate intervals, it is decidable in $O(2^g \text{poly}(n))$ time. In contrast, one hard band, treewidth one, or radius two alone does not remove NP-hardness. Fourth, exact common-realization semantics extends from independent interval boxes to correlated rational-polyhedral uncertainty. For a fixed deterministic trace, every query cell is rational polyhedral and is testable by linear programming, yielding polynomial-bit rational witnesses. Finally, we define a finite multichannel and spatiotemporal product theory in which scalar EF-FG layers are coupled by one joint polyhedral realization and explicit finite cross-layer predicates; deterministic evaluation, self-duality, exact certification, and NP-hardness transfer layerwise. The resulting package gives an exact universality theorem, a local/global complexity boundary, parameterized refinements, and correlated/layered extensions in one closed finite framework.

Keywords. scalar fields; foreground/background; merge trees; common realization; simplicial complexes; NP-completeness; fixed-parameter tractability; polyhedral uncertainty; multichannel data; time-varying scalar fields.

1 Introduction

The foundational EF-FG theory introduced a self-dual foreground/background framework for finite scalar fields, separating two projections that should not be conflated: the spatial-frame projection SF_x , which records carrier, gate, route, terminal, and residual-frame information, and the polarity-core ownership projection PO_x , which records observer-relative bright/dark saliency ownership. Its uncertainty semantics are exact in the following sense: all quantities appearing in a claimed state must arise from one common scalar realization rather than from independently optimized marginal envelopes. The first paper also proved a sharp local/global separation for

the ownership projection, including NP-completeness of a q-band owner-scheduling kernel, its exact lift to interval-valued radius-two trees, and NP-completeness of Possible Background Polarity-Core Ownership at fixed parameters.

The foundational paper identified five explicit theorem-sized directions: the complexity of the full spatial-frame projection, characterization of finite common-realization complexes, parameterized complexity, exact semantics beyond independent interval boxes, and finite multi-channel/spatiotemporal extensions. This paper closes all five within explicit finite scopes. The central theme is that the same common-realization principle controls seemingly different questions: spatial-frame synchronization, realization topology, owner-order combinatorics, correlated uncertainty, and cross-layer consistency.

The strongest new structural result is a universality theorem. Protected obligation families define downward-closed common-realization complexes. Earlier examples realized all simplex boundaries. Here we prove the exact classification

$$\boxed{\{K_{\text{raw}}(P)\} = \{K_{\sigma}(P)\} = \{\text{all finite simplicial complexes}\}.$$

Thus no nontrivial finite simplicial-complex obstruction survives at the level of realizability classes.

The strongest new complexity result concerns the spatial projection. We define a native terminal-mixed-mask decision problem and then a complete finite categorical spatial-frame realization problem. The latter is NP-complete even when the graph is bipartite, the radius is at most three, the seed threshold is fixed, uncertainty is confined to identical boundary-terminal intervals, and all other scalar values come from a constant rational vocabulary. This hardness mechanism is distinct from q-band ownership scheduling: it uses terminal exposure, gate floors, strong incidence, midpoint signs, and one common realization.

The paper then refines the ownership hardness by identifying what actually makes the q-band search explode. A bounded ambiguity core in elder order yields factorial FPT, while a bounded number of uncertain attachment gates yields subset FPT. These positive parameters are orthogonal to shallow graph topology: the hard family already has one hard band, treewidth one, and radius two.

Finally, the exact semantics is extended in two directions. Correlated rational-polyhedral uncertainty replaces the independent box by a compact or extended linear constraint system. Fixing a deterministic EF–FG trace linearizes every min/max and comparison branch, so each trace cell is a rational polyhedron and fixed-trace feasibility is an LP. A finite layered construction then treats channels or time slices as ordinary scalar EF–FG layers coupled by one joint realization and explicit finite cross-layer predicates. This is deliberately not a vector-valued merge-tree theory; it is an exact product/coupling theorem for the scalar framework.

Relation to prior work. Merge trees, component trees, and trees of shapes provide established scalar-field and morphological summaries [2, 3, 7, 8]. Uncertainty in scalar-field topology has been studied through mandatory critical points and statistical/ensemble summaries [5, 6, 10, 11]. Multivariate topology uses genuinely multivariate objects such as Reeb spaces and related simplifications [12], while temporal scalar-field analysis includes merge-tree alignment, tracking, and time-varying contour topology [13, 14]. The present paper does not claim novelty for these primitives. Its claims concern the specific EF–FG common-realization decision problems, the finite-complex universality theorem inside EF–FG semantics, the parameterized boundary of the q-band kernel, and the exact polyhedral/layered transfer theorems proved below.

Paper organization. Section 2 fixes the notation needed from EF–FG I. Section 3 proves global spatial-frame NP-completeness. Section 4 proves raw and canonical finite-complex universality. Section 5 gives the parameterized boundary. Section 6 develops correlated rational-polyhedral

uncertainty. Section 7 gives the finite multichannel/spatiotemporal transfer. Section 8 records the resulting closed theorem package.

2 Preliminaries from EF–FG I

We use the scalar-field and observer semantics of the foundational EF–FG paper. Only the ingredients required for the new theorems are recalled here.

Let $G = (V, E)$ be a finite graph and $F : V \rightarrow \mathbb{R}$ a scalar field. For vertices u, v , the maximin connectivity is

$$\kappa_F(u, v) = \max_{\pi: u \rightsquigarrow v} \min_{w \in \pi} F(w).$$

A resolved bright birth atom has a birth value, an elder order, and an observer attachment. Dark quantities are defined by applying the same construction to $-F$. The observer-record theorem from EF–FG I reduces pointwise polarity-core saliency to a record chain: a finite birth is relevant exactly when its observer attachment strictly raises the running elder pressure.

For spatial-frame structure, fix the selected finite birth atoms. For a seed p define

$$\kappa_p(v) = \kappa_F(p, v), \quad \beta(v) = \max_r \kappa_r(v), \quad W(v) = \{p : \kappa_p(v) = \beta(v)\}.$$

The rooted co-winner basin B_p is the component containing p in $\{v : p \in W(v)\}$. If p, q are carrier-adjacent, their gate floor is

$$s(p, q) = \max_{v \in B_p \cap B_q} \beta(v).$$

For a boundary terminal $t \in B_p$, the terminal gate is

$$s_\partial(p, t) = \kappa_{F|B_p}(p, t).$$

For an observer x , the bright midpoint margin against a gate floor s is

$$\mu_{p,q,x}^+ = 2F(x) - F(p) - s,$$

with the dark analogue obtained from $-F$. Weak incidence means that at least one admissible top route passes through x , while strong incidence means that every admissible top route does.

Under uncertainty, a claimed state must arise from one common realization. For a realization family $\mathcal{P}$ and a finite query Q ,

$$\text{Possible}_{\mathcal{P}}(Q) \iff \exists F \in \mathcal{P}, \exists \rho : Q(F, \rho),$$

where ρ denotes the finite elder/tie resolution. Robust semantics use universal quantification over the same common-realization space.

For polarity-core complexity we also use the q-band kernel. At $\theta = 1/2$, a hard-band state is (h, x) and event i has parameters (h_i, A_i, B_i) . With

$$s_i = \min\{h, h_i\},$$

owner use is legal exactly when

$$B_i < x, \quad \frac{x - s_i}{2} \leq A_i,$$

and the transition is

$$(h, x) \mapsto \left(s_i, \max \left\{ B_i, \frac{x - s_i}{2} \right\} \right).$$

For a fixed admissible event order, the prefix-minimum theorem of EF–FG I evaluates existential completion in linear time. The NP-completeness therefore lies in choosing a successful order among incomparable one-use owner events, not in evaluating a proposed order.

3 Global spatial-frame realization complexity

The ownership hardness of EF-FG I does not determine the complexity of the spatial-frame projection. We now classify a natural finite categorical realization problem for that projection.

3.1 Terminal mixedness

For one polarity and observer x , let

$$M_T(x) = (W^+, W^-, S^+, S^-)$$

be the terminal peak-side mask. The bit W^+ is one iff some weakly incident exposed terminal sector has positive midpoint margin; W^- is one iff some such sector has nonpositive margin; S^+ and S^- use strong incidence.

Definition 3.1 (Possible Terminal-Mixed Mask (PTMM)). The input consists of a finite graph, rational interval scalar values, a boundary-terminal set T , an observer set X , and a rational seed threshold. The question is whether one common scalar realization and one elder/tie resolution satisfy

$$M_T(x) = (1, 1, 1, 1) \quad \text{for every } x \in X.$$

Theorem 3.2 (Terminal-mask hardness). *PTMM is NP-complete. NP-hardness holds simultaneously when the graph is bipartite, has radius two about a specified hub, the seed threshold is fixed at 2, uncertainty occurs only on boundary terminals, every uncertain terminal has interval $[5, 7]$, and every other scalar value belongs to $\{0, 8, 10, 20\}$.*

Proof. Reduce from Monotone NAE-3-SAT-E4 [9]. Let h be a hub with $F(h) = 0$ and let e be an essential peak with $F(e) = 20$. For each clause c , create an observer x_c with value 8 and a clause peak p_c with value 10, with edges

$$h - x_c - p_c.$$

For each Boolean variable i , create one shared boundary terminal z_i with

$$F(z_i) \in [5, 7],$$

and join x_c to z_i iff variable i occurs in clause c .

The graph is bipartite with hub radius two. The selected finite seeds are exactly the clause peaks. At x_c the unique rooted seed is p_c : its score is 8, while any other clause peak reaches x_c through a bottleneck at most 7. A terminal z_i belongs to B_{p_c} exactly when $i \in c$. In that case the path $p_c - x_c - z_i$ lies in the basin, the terminal gate is

$$s_{\partial}(p_c, z_i) = F(z_i),$$

and x_c is strongly incident because no basin route can bypass it.

Therefore the terminal midpoint margin is

$$\mu_{c,i} = 2F(x_c) - F(p_c) - s_{\partial}(p_c, z_i) = 16 - 10 - F(z_i) = 6 - F(z_i).$$

Hence a clause observer has mask 1111 exactly when its three terminals contain at least one value below 6 and at least one value at least 6. A NAE assignment gives such a realization by mapping the two truth values to 5 and 7. Conversely, any successful realization induces a Boolean assignment by the threshold $F(z_i) < 6$ versus $F(z_i) \geq 6$, and every clause is NAE-satisfied.

NP membership follows from a polynomial-bit rational realization certificate together with a polynomial-size deterministic comparison/maximin trace. Once the trace is fixed, all scalar requirements are rational linear equalities and strict/non-strict inequalities; a nonempty trace cell has a polynomial-bit rational point. Verification of the deterministic spatial mask is polynomial. $\square$

3.2 Complete categorical spatial-frame realization

PTMM constrains a native coordinate block but is still a cylinder query. We now lift it to a fully prescribed finite categorical state.

For one polarity define

$$\text{CSF}^\sigma(x) = (\rho^\sigma(x), M_{\text{aug}}^\sigma(x), N^\sigma(x), R^\sigma(x)),$$

where ρ records unique-rooted/co-winner/no-finite-root status, M_{aug} is the complete competitor+terminal four-bit mask, N is the exact-neutral flag, and R is the categorical residual-terminal state. The complete self-dual state is

$$\text{CSF}(x) = (\text{CSF}^+(x), \text{CSF}^-(x)).$$

Exact real-valued gate/cap coordinates are deliberately excluded from this finite quotient.

Definition 3.3 (Complete Categorical Spatial-Frame Realization (CCSFR)). Given an interval-valued graph, boundary terminals, observer set X , fixed seed threshold, and one prescribed finite categorical target τ_x for each observer, decide whether one common realization and one elder/tie resolution satisfy

$$\text{CSF}(x) = \tau_x \quad (x \in X).$$

Theorem 3.4 (Complete spatial-state hardness). *CCSFR is NP-complete. Hardness holds on bipartite graphs of radius at most three with fixed seed threshold 2, uncertainty only on identical terminal intervals $[5, 7]$, and all fixed scalar values in the constant rational set*

$$\{-8, 0, 5/2, 4, 8, 19/2, 12, 20\}.$$

Proof. Use the same Boolean coordinates $z_i \in [5, 7]$ in two mirror clause gadgets. In the bright gadget choose

$$F(x_c^+) = 8, \quad F(p_c^+) = 19/2.$$

The active terminal margin is

$$\mu_{c,i}^+ = \frac{13}{2} - z_i.$$

Thus the required bright nonpositive bit exists exactly when some clause variable has $z_i > 6.5$ once exact neutrality is separately excluded.

For the dark mirror use inversion constant 12 with

$$F(x_c^-) = 4, \quad F(p_c^-) = 5/2.$$

In dark-normalized coordinates the corresponding margin is

$$\mu_{c,i}^- = z_i - \frac{11}{2},$$

so the required dark nonpositive bit exists exactly when some clause variable has $z_i < 5.5$.

The remaining categorical coordinates are frozen throughout $z_i \in [5, 7]$: the selected seed family is unchanged; same-polarity query roots remain unique; opposite-polarity query roots remain co-winners/nonunique; the observer stays on the forced peak side of each active terminal gate, so the residual coordinate is empty; and no unintended neutral sector appears under the prescribed target. Hence each bright observer can be prescribed the full target

$$((\text{unique}, 1111, 0, \text{empty}), (\text{co-winner}, 0000, 0, \text{empty}))$$

and each dark observer the mirror target.

If the NAE instance is satisfiable, map the two truth values to 7 and 5. Every clause then has one value above 6.5 and one below 5.5, so all full targets are realized. Conversely, a realization of every full target gives in each clause one $z_i > 6.5$ and one $z_j < 5.5$. Assign these two ranges opposite Boolean values and assign middle values arbitrarily. Every clause already has both truth values and is NAE-satisfied.

Membership in NP uses the same polynomial trace-cell argument as Theorem 3.2. $\square$

Corollary 3.5. *Exact arbitrary querying of the richer numerical full spatial-frame realization relation is NP-hard.*

The mechanism of Theorems 3.2–3.4 is independent of polarity-core owner scheduling. It uses terminal exposure, gate floors, route incidence, midpoint signs, and synchronization through one common scalar realization.

4 Finite-complex realization universality

Let P be a finite family of named protected strict observer-record obligations. As in EF–FG I, define

$$K_{\text{raw}}(P) = \{A \subseteq P : \text{one admissible raw realization satisfies every obligation in } A\}$$

and let $K_\sigma(P)$ be the corresponding complex under the canonical maximal-safe rule. Both are finite simplicial complexes.

Earlier constructions produced all simplex boundaries. We now classify the realizability classes completely.

4.1 Facet-selector construction

Let K be a finite simplicial complex on vertices $1, \dots, n$ with facets

$$F_1, \dots, F_m.$$

Create one observer x with $F(x) = 4$. For every facet create a selector gate g_j with uncertain value

$$y_j := F(g_j) \in [2, 3].$$

Choose rational caps

$$2 < c_1 < \dots < c_n < 3.$$

For every complex vertex i , create a protected peak p_i and a private elder blocker q_i . Their maximin attachments to x are arranged so that, for every selector realization y ,

$$a_i(y) = \kappa(p_i, x) = \min \left(c_i, \max_{j: i \in F_j} y_j \right),$$

and

$$b_i(y) = \kappa(q_i, x) = \min \left(c_i, \max \left(2, \max_{j: i \notin F_j} y_j \right) \right).$$

Fix an essential elder with observer attachment 1 and choose finite births in $(6, 7)$ with elder order

$$q_1 > p_1 > q_2 > p_2 > \dots > q_n > p_n.$$

The graph is bipartite and has radius at most three about the observer.

Lemma 4.1 (Maximizer-facet soundness). *For every selector realization, every protected peak that is a strict record belongs to every facet attaining the global maximum selector value. In particular the protected record set is a face of K .*

Proof. If p_i is a strict record against its private blocker, then $a_i > b_i$. Since the same cap c_i is applied to both sides, this forces

$$\max_{j:i \in F_j} y_j > \max \left(2, \max_{j:i \notin F_j} y_j \right).$$

Let F_{j^*} be any facet with globally maximal y_{j^*} . If $i \notin F_{j^*}$, the right-hand maximum is at least y_{j^*} , while the left-hand maximum cannot exceed it, contradicting the strict inequality. Thus every strict protected record lies in every maximizing facet. $\square$

Lemma 4.2 (Facet completeness). *For every facet F_t there is a realization whose protected strict-record set is exactly F_t .*

Proof. Set

$$y_t = 3, \quad y_j = 2 \quad (j \neq t).$$

Then for $i \in F_t$ we obtain $a_i = c_i$ and $b_i = 2$, while for $i \notin F_t$ the blocker is not below the protected attachment. With increasing caps and the fixed elder birth order, exactly the p_i with $i \in F_t$ are strict observer records. $\square$

Theorem 4.3 (Raw universality). *For every finite simplicial complex K there exists a finite interval-valued EF-FG graph and protected obligation family P such that*

$$K_{\text{raw}}(P) \cong K.$$

The construction may be chosen with one observer, a bipartite graph of radius at most three, uncertainty only on facet selectors, and identical selector interval $[2, 3]$.

Proof. Lemma 4.1 gives $K_{\text{raw}}(P) \subseteq K$. Lemma 4.2 realizes every facet, hence every subface, so $K \subseteq K_{\text{raw}}(P)$. $\square$

4.2 Canonical freezing

The raw construction becomes canonical because its geometry already forces every relevant attachment once the selector vector is fixed.

Lemma 4.4 (Forced-safe canonical fiber). *Fix one selector realization y . For every protected peak and private blocker, the feasible attachment fiber is the singleton $\{a_i(y)\}$ or $\{b_i(y)\}$. Moreover every such attachment lies strictly below the canonical hard-band safety ceiling at $\theta = 1/2$ and $H = 0$.*

Proof. The maximin construction itself fixes the attachment values. Every attachment is below 3, whereas all finite births exceed 6 and the essential elder pressure is 1. Hence the canonical safety ceiling satisfies

$$\frac{B + D}{2} > \frac{6 + 1}{2} = 3.5 > 3.$$

Thus every geometrically forced attachment is strictly safe. The maximal-safe canonical section has no alternative value to choose and therefore equals the forced raw attachment. $\square$

Theorem 4.5 (Canonical universality). *For every finite simplicial complex K there exists a finite interval-valued EF-FG instance with*

$$K_{\sigma}(P) \cong K.$$

Consequently

$$\{K_{\text{raw}}(P)\} = \{K_{\sigma}(P)\} = \{\text{all finite simplicial complexes}\}.$$

Proof. By Lemma 4.4, realization by realization the canonical and raw elder scans coincide on the protected family. The proof of Theorem 4.3 therefore transfers verbatim. $\square$

Remark 4.6. The equality in Theorem 4.5 is an equality of realizability classes. It does not assert $K_{\text{raw}}(P) = K_{\sigma}(P)$ for every fixed EF–FG instance. The canonical/raw distinction remains genuine when an instance has nontrivial intermediate attachment freedom.

5 Parameterized complexity of the ownership kernel

The q-band kernel is NP-complete, but fixed-order evaluation is linear. We now identify two natural parameters that measure the remaining global choice.

5.1 Elder-order ambiguity

For event birth intervals

$$I_i = [\ell_i, u_i],$$

define forced elder precedence

$$i \prec_B j \iff \ell_i > u_j.$$

Let C_B be the set of events incomparable with at least one other event under this relation, and put

$$k = |C_B|.$$

Lemma 5.1 (Linear extensions are exactly realizable elder orders). *A total event order is realizable by scalar choices from the birth intervals iff it is a linear extension of $\prec_B$.*

Proof. Necessity is immediate. For sufficiency, write the proposed elder-to-younger order as $i_1, \dots, i_n$ and construct birth values backwards. Set the last value to any point in its interval and recursively choose

$$b_{i_t} = \max\{\ell_{i_t}, b_{i_{t+1}}\}.$$

If this exceeded u_{i_t} , then $b_{i_{t+1}} > u_{i_t}$, and some later event would be forced elder to i_t , contradicting the linear-extension property. Ties are resolved in the proposed order. $\square$

Lemma 5.2 (Ambiguity-core enumeration). *The events outside C_B form one total chain and are comparable with every event. Fixing a linear order of the k ambiguity-core events determines at most one full linear extension. Hence the number of admissible elder orders is at most $k!$.*

Proof. If two events outside C_B were incomparable, each would belong to C_B . Thus the outside set is a chain. For each core event, its comparisons with the outside chain determine a unique insertion slot. A chosen core permutation is either inconsistent with the internal precedence constraints or yields one full extension. $\square$

Theorem 5.3 (Order-ambiguity FPT). *Q-band completion is fixed-parameter tractable in k :*

$$T(n, k) = O(k! \text{poly}(n)).$$

For one hard band the bound is $O(k!n)$ after the admissible orders have been generated.

Proof. Enumerate the at most $k!$ core permutations, reconstruct the corresponding full linear extension when one exists, and evaluate that fixed order by the EF–FG I prefix-minimum scan. $\square$

5.2 Uncertain attachment gates

For event $e_i = (h_i, A_i, B_i)$, the corresponding physical attachment interval is

$$[q - A_i, q - B_i].$$

Call it uncertain iff $A_i > B_i$, and let

$$g = \#\{i : A_i > B_i\}.$$

Lemma 5.4 (Deterministic-gate reset). *If $A_i = B_i = b_i$ and event i is owner-legal, then the post-event deficit is exactly*

$$x' = b_i.$$

Proof. Owner legality gives $(x - s_i)/2 \leq b_i$, so

$$x' = \max\{b_i, (x - s_i)/2\} = b_i.$$

□

Lemma 5.5 (Deterministic-gate chain). *If two deterministic-gate events satisfy $B_i > B_j$, then any legal owner word using both must place i before j . Equal- B deterministic events cannot both be owner uses.*

Proof. If j were used first, Lemma 5.4 would leave $x = B_j$, after which event i would require $B_i < x = B_j$, impossible. For equal B , the first reset makes the second fail the strict owner-live inequality. □

Define state dominance by

$$(h_1, x_1) \succeq (h_2, x_2) \iff h_1 \geq h_2, \quad x_1 \leq x_2.$$

A dominating state can simulate every successful future continuation of the dominated state, possibly skipping an event that has already become owner-dead.

Theorem 5.6 (Uncertain-gate FPT). *Q -band completion is fixed-parameter tractable in g :*

$$\boxed{T(n, g) = 2^g \text{poly}(n).}$$

A straightforward exact dynamic program runs in $O(2^g g n^3)$ arithmetic operations.

Proof. Sort the deterministic-gate events into the forced chain $d_1, \dots, d_m$ from Lemma 5.5, and let U be the g uncertain-gate events. Use DP cells

$$DP[j, S], \quad 0 \leq j \leq m, \quad S \subseteq U,$$

storing the Pareto frontier of reachable hard states after deciding the first j deterministic events and using exactly the uncertain owner events in S . From each state one may skip d_{j+1} , use it if legal, or use any unused uncertain event if legal. Every legal owner word is represented because its deterministic subsequence must respect the fixed chain.

Reachable headroom values belong to the finite set $\{h_0, h_1, \dots, h_n\}$. For a fixed headroom, only the smallest deficit survives dominance pruning, so each frontier has at most $n + 1$ states. There are $(m + 1)2^g$ cells and at most $g + 1$ owner choices per state, yielding the displayed FPT bound. □

Corollary 5.7 (Parameterized boundary). *On the q -band/radius-two family carrying the ownership hardness theorem:*

$$\boxed{\text{small } k \text{ or small } g \implies \text{FPT.}}$$

By contrast, band count, graph treewidth, and graph radius alone are para-NP-hard parameters: NP-hardness already holds at one hard band, treewidth one, and radius two.

Proof. The positive statements are Theorems 5.3–5.6 and transfer through the exact radius-two physical lift and dark-anchor lift of EF–FG I. The negative statements follow immediately from the restrictions of the existing NP-completeness construction. $\square$

6 Correlated rational-polyhedral uncertainty

Independent interval boxes are useful but cannot express shared calibration, conservation, monotonicity, or other cross-vertex correlations. We now replace the box by an exact correlated class.

Definition 6.1 (Rational-polyhedral realization model). Let $F \in \mathbb{R}^V$ be the scalar field and $z \in \mathbb{R}^r$ auxiliary latent variables. A realization model is given in extended form by

$$\widehat{\mathcal{P}} = \{(F, z) : AF + Bz \leq c, CF + Dz = e\},$$

with rational data. The admissible scalar fields are

$$\mathcal{P} = \text{proj}_F \widehat{\mathcal{P}}.$$

The class contains interval boxes, shared offsets, linear calibration models, Lipschitz constraints, monotonicity constraints, affine conservation laws, and finite intersections of rational halfspaces and hyperplanes.

Theorem 6.2 (Polyhedral trace theorem). *Fix a finite categorical or rational-threshold EF–FG query Q . There exists a finite family of discrete deterministic traces $\mathfrak{T}_Q$ such that the realization set of Q is a finite union of projections of relatively open rational polyhedra:*

$$\{F \in \mathcal{P} : \exists \rho \ Q(F, \rho)\} = \bigcup_{\tau \in \mathfrak{T}_Q} \text{proj}_F \widehat{\mathcal{P}}_\tau.$$

For a fixed trace τ , feasibility of $\widehat{\mathcal{P}}_\tau$ is decidable in polynomial time by rational linear programming.

Proof. The deterministic EF–FG evaluator is a polynomial-time finite comparison program built from scalar coordinates, rational affine combinations, min/max operations, comparisons, graph searches whose active subgraphs are comparison-determined, and finite elder/tie choices. A trace records every comparison branch, chosen min/max attainer, elder/tie decision, and resulting query branch. Once the trace is fixed, every numerical intermediate value is a rational affine function of F . Trace compatibility is therefore a rational system

$$A_\tau F \leq b_\tau, \quad C_\tau F = d_\tau, \quad S_\tau F < t_\tau.$$

Intersecting with the extended uncertainty model gives a relatively open rational polyhedron.

For strict inequalities introduce one common slack $0 \leq \delta \leq 1$ and replace $r < s$ by $r + \delta \leq s$. Maximizing δ yields strict feasibility iff the optimum is positive. This is a rational LP of polynomial size for a fixed trace. $\square$

Corollary 6.3 (Polynomial rational witnesses). *Every feasible existential finite EF–FG query under rational-polyhedral uncertainty has a polynomial-bit rational common realization and polynomial-size finite trace/tie certificate. Hence such possible queries lie in NP whenever deterministic verification is polynomial. Their universal robust complements lie in coNP.*

Proof. A feasible strict rational trace LP with positive slack has a polynomial-bit basic feasible solution by standard determinant bounds for rational linear systems. Verification replays the deterministic trace and checks the joint polyhedral constraints exactly. $\square$

Proposition 6.4 (Self-duality). *If $\mathcal{P}$ is rational-polyhedral, then*

$$-\mathcal{P} = \{-F : F \in \mathcal{P}\}$$

is rational-polyhedral, and the realization-wise EF-FG duality

$$\mathcal{E}(-F) = \mathcal{D}(\mathcal{E}(F))$$

extends exactly to possible and robust polyhedral semantics.

Proof. Substitute $F = -F'$ in the rational extended formulation and use the deterministic inversion theorem layerwise. $\square$

Corollary 6.5 (Complexity transfer). *All interval-box NP-hardness and NP-completeness results proved in EF-FG I and Section 3 remain valid for rational-polyhedral uncertainty.*

Proof. An interval box is a rational polytope. NP membership follows from Corollary 6.3. $\square$

Remark 6.6 (Scope). The old interval-cactus dynamic programs exploit coordinatewise independence and local separator factorization. Arbitrary dense linear correlations can couple distant blocks, so we do not claim that those specific DPs remain polynomial. The generic exact replacement is finite trace enumeration with LP feasibility.

7 Finite multichannel and spatiotemporal transfer

We now extend the scalar theory to a finite family of coupled scalar layers without imposing a vector-valued total order.

Definition 7.1 (Finite layered EF-FG). Let $\Lambda = \{1, \dots, L\}$ be a finite layer set. Layer λ has graph $G_\lambda = (V_\lambda, E_\lambda)$, scalar field F_λ , and ordinary deterministic EF-FG output $\mathcal{E}_\lambda(F_\lambda, \rho_\lambda)$. The uncoupled product output is

$$\mathcal{E}_\Lambda(F, \rho) = (\mathcal{E}_1, \dots, \mathcal{E}_L).$$

All layer scalar coordinates and optional latent variables are jointly constrained by one rational extended polyhedron.

A finite cross-layer specification Γ may contain categorical relation atoms

$$(s_{\lambda,a}, s_{\mu,b}) \in R$$

for an explicitly supplied finite relation R , and rational affine atoms in named numerical observables,

$$\sum_j q_j y_j \bowtie b, \quad \bowtie \in \{<, \leq, =, \geq, >\}.$$

For time layers, finite correspondence relations between named observers/features may be supplied as part of the input. No automatic feature-identity theorem is assumed.

Theorem 7.2 (Deterministic layered transfer). *For fixed realizations and elder/tie resolutions, finite layered EF-FG is evaluable in*

$$\boxed{O\left(\sum_{\lambda=1}^L T_\lambda + |\Gamma|\right)},$$

where T_λ is the deterministic evaluation time of layer λ .

Proof. Evaluate every scalar layer independently and then evaluate the finite cross-layer predicates. $\square$

Theorem 7.3 (Joint layered trace theorem). *Fix one deterministic trace for each layer and one Boolean branch of every predicate in Γ . The joint realization cell is a relatively open rational polyhedron in the concatenated layer and latent variables. Fixed joint-trace feasibility is rational LP feasibility, and every feasible possible query has a polynomial-bit rational joint witness.*

Proof. Apply Theorem 6.2 to each layer. On a fixed layer trace every named numerical observable is rational affine in that layer's scalar coordinates. The cross-layer affine atoms remain affine in the concatenated vector; categorical relation atoms become fixed truth values. One common slack handles all strict inequalities. Corollary 6.3 then gives a polynomial-bit rational joint solution. $\square$

Theorem 7.4 (Layered self-duality). *Under simultaneous inversion*

$$\mathfrak{I}(F_1, \dots, F_L) = (-F_1, \dots, -F_L),$$

finite layered EF-FG satisfies

$$\boxed{\mathcal{E}_\Lambda(\mathfrak{I}F) = \mathcal{D}_\Lambda(\mathcal{E}_\Lambda(F))},$$

with $\mathcal{D}_\Lambda = \mathcal{D}_1 \times \dots \times \mathcal{D}_L$. If Γ is transformed by the corresponding duality action, possible and robust statements are preserved; if Γ is duality-invariant, the entire model is self-dual.

Proof. Apply the scalar inversion theorem to every layer and the linear substitution $F \mapsto -F$ to the joint polyhedral model. $\square$

Theorem 7.5 (Independent-product factorization). *Suppose*

$$\mathcal{P} = \prod_{\lambda=1}^L \mathcal{P}_\lambda$$

and the query is a conjunction $Q = \bigwedge_\lambda Q_\lambda$ of layer-local predicates with no cross-layer atom. Then

$$\text{Possible}_{\mathcal{P}}(Q) \iff \bigwedge_{\lambda} \text{Possible}_{\mathcal{P}_\lambda}(Q_\lambda),$$

and

$$\text{Robust}_{\mathcal{P}}(Q) \iff \bigwedge_{\lambda} \text{Robust}_{\mathcal{P}_\lambda}(Q_\lambda).$$

Proof. For possible semantics, combine independent layer witnesses into their product realization; the converse is projection. The robust statement follows from universal quantification over a Cartesian product. $\square$

Theorem 7.6 (Two-layer replication transfer). *Let Q be any single-layer EF-FG decision query. Construct two layer copies and impose exact equality*

$$F_2(v) = F_1(v)$$

for corresponding scalar coordinates. Require the same target query in both layers. Then the single-layer and two-layer instances are equivalent. Hence every NP-hard single-layer EF-FG query remains NP-hard already with exactly two channels or exactly two time slices; when the single-layer query is in NP, the replicated version is NP-complete.

Proof. A single-layer witness can be copied to both layers with the same resolution. Conversely, a successful two-layer realization projects to a successful realization of either layer. The equality coupling is rational-polyhedral and the transformation is polynomial. $\square$

A temporal state-persistence requirement is a special categorical relation. For example, if S is a finite queried state space, the relation

$$R_{\text{id}} = \{(s, s) : s \in S\}$$

enforces exact state persistence across adjacent frames. Theorem 7.6 remains valid with this explicit temporal consistency predicate.

8 Synthesis: the closed theorem package

The new results complete the theorem-sized directions left by EF–FG I.

Theorem 8.1 (EF–FG II completion theorem). *Within the finite scopes defined in this paper, the following statements hold.*

- (i) *Full finite categorical spatial-frame realization is NP-complete on severely restricted shallow bipartite graphs (Theorem 3.4).*
- (ii) *Every finite simplicial complex occurs as a raw EF–FG common-realization complex and as a canonical EF–FG common-realization complex (Theorems 4.3–4.5).*
- (iii) *Q-band ownership completion is FPT both in elder-order ambiguity k and in uncertainty gate count g , while one hard band, treewidth one, and radius two alone do not remove NP-hardness (Corollary 5.7).*
- (iv) *Exact common-realization semantics extends from independent interval boxes to correlated rational-polyhedral uncertainty with polyhedral trace cells, LP fixed-trace feasibility, polynomial-bit rational witnesses, and preserved self-duality (Theorem 6.2 and Proposition 6.4).*
- (v) *Finite multichannel and spatiotemporal EF–FG admits exact joint-realization semantics, deterministic product evaluation, polyhedral trace certification, self-duality, factorization in the uncoupled case, and two-layer complexity transfer (Theorems 7.2–7.6).*

These statements sharpen the original local/global picture. Global difficulty does not come from deep graph topology: ownership hardness survives on radius-two trees, while spatial-frame hardness survives on shallow bipartite graphs. At the same time, difficulty is not uniform. Order ambiguity and gate uncertainty expose finite-parameter routes back to exact tractability. The realization-complex theorem shows that structural compatibility is maximally expressive at finite scale, while the polyhedral and layered theorems show that exact common-realization semantics is stable under a substantial enlargement of the uncertainty and data model.

The scope boundaries are part of the theorem package. The finite layered theory is not a multivariate Reeb-space theory, and the polyhedral theorem does not assert preservation of interval-cactus factorization under dense correlations. Likewise, the equality of raw and canonical realizability classes does not identify the two complexes for each individual instance. Stating these distinctions explicitly keeps the sequel within the semantics actually proved.

9 Conclusion

This paper turns the open theorem program of EF–FG I into a closed second-stage theory. The spatial-frame projection now has a direct NP-completeness classification at the complete finite categorical level. Protected common-realization complexes are universal among finite simplicial complexes under both raw and canonical semantics. The ownership hardness kernel admits two exact FPT escape parameters, distinguishing genuine order/gate ambiguity from shallow

graph structure. Common-realization semantics extends exactly to correlated rational-polyhedral uncertainty, and finite scalar layers can be coupled into multichannel or spatiotemporal models without abandoning self-duality or exact certification.

The combined picture is therefore not simply that global EF–FG questions are hard. Rather, the theory identifies three complementary regimes: local or fixed-order calculations with direct polynomial algorithms; globally synchronized realization problems that are NP-complete even on shallow graph families; and intermediate instances whose global uncertainty is concentrated in a small ambiguity core and is fixed-parameter tractable. At the structural level, finite compatibility patterns are fully universal. At the modeling level, exact semantics survives correlated and layered coupling by remaining faithful to one common realization before projection.

Statements and declarations

Author contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, software, validation, visualization, writing–original draft, and writing–review and editing.

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Data and code availability. The paper is mathematical and uses no empirical dataset. All definitions, constructions, and proofs needed for the stated mathematical results are contained in the manuscript. A reference implementation and executable regression checks are auxiliary validation tools and are not premises of the proofs. No external dataset is required to reproduce the theoretical results.

Ethics approval and consent. Not applicable.

Consent for publication. Not applicable.

References

- [1] A. E. Bütün, *Extrema–Frame Foreground–Background Theory: Self-Dual Structure, Exact Common-Realization Semantics, and Ownership Complexity*, Zenodo, 2026. doi:10.5281/zenodo.22944661.
- [2] P. Monasse and F. Guichard, Fast Computation of a Contrast-Invariant Image Representation, *IEEE Transactions on Image Processing* 9(5) (2000), 860–872. doi:10.1109/83.841532.
- [3] T. Liu, M. Seyedhosseini, and T. Tasdizen, Image Segmentation Using Hierarchical Merge Tree, *IEEE Transactions on Image Processing* 25(10) (2016), 4596–4607. doi:10.1109/TIP.2016.2592704.
- [4] A. G. Silva and R. de Alencar Lotufo, Efficient Computation of New Extinction Values from Extended Component Tree, *Pattern Recognition Letters* 32(1) (2011), 79–90. doi:10.1016/j.patrec.2010.07.019.
- [5] D. Günther, J. Salmon, and J. Tierny, Mandatory Critical Points of 2D Uncertain Scalar Fields, *Computer Graphics Forum* 33(3) (2014), 31–40. doi:10.1111/cgf.12359.
- [6] L. Yan, Y. Wang, E. Munch, E. Gasparovic, and B. Wang, A Structural Average of Labeled Merge Trees for Uncertainty Visualization, *IEEE Transactions on Visualization and Computer Graphics* 26(1) (2020), 832–842. doi:10.1109/TVCG.2019.2934242.
- [7] T. Géraud, N. Boutry, S. Crozet, E. Carlinet, and L. Najman, A Proof of the Tree of Shapes in n-D, arXiv:2206.05109, 2022. doi:10.48550/arXiv.2206.05109.

- [8] J. Mendes Forte, N. Passat, A. Shimizu, and Y. Kenmochi, Consistent Connected Operators Based on Trees of Shapes, *SIAM Journal on Imaging Sciences* 18(4) (2025), 2547–2579. doi:10.1137/25M174704X.
- [9] A. Darmann and J. Döcker, On a simple hard variant of Not-All-Equal 3-Sat, *Theoretical Computer Science* 815 (2020), 147–152. doi:10.1016/j.tcs.2020.02.010.
- [10] T. M. Athawale, D. Maljovec, L. Yan, C. R. Johnson, V. Pascucci, and B. Wang, Uncertainty Visualization of 2D Morse Complex Ensembles Using Statistical Summary Maps, *IEEE Transactions on Visualization and Computer Graphics* 28(4) (2022), 1955–1966. doi:10.1109/TVCG.2020.3022359.
- [11] T. M. Athawale, Z. Wang, D. Pugmire, K. Moreland, Q. Gong, S. Klasky, C. R. Johnson, and P. Rosen, Uncertainty Visualization of Critical Points of 2D Scalar Fields for Parametric and Nonparametric Probabilistic Models, *IEEE Transactions on Visualization and Computer Graphics* 31(1) (2025), 108–118. doi:10.1109/TVCG.2024.3456393.
- [12] A. Chattopadhyay, H. Carr, D. Duke, Z. Geng, and O. Saeki, Multivariate Topology Simplification, *Computational Geometry* 58 (2016), 1–24. doi:10.1016/j.comgeo.2016.05.006.
- [13] W. Köpp and T. Weinkauff, Temporal Merge Tree Maps: A Topology-Based Static Visualization for Temporal Scalar Data, *IEEE Transactions on Visualization and Computer Graphics* 29(1) (2023), 1157–1167. doi:10.1109/TVCG.2022.3209387.
- [14] M. Li, X. Yan, L. Yan, T. Needham, and B. Wang, Flexible and Probabilistic Topology Tracking With Partial Optimal Transport, *IEEE Transactions on Visualization and Computer Graphics* 31(10) (2025), 7951–7969. doi:10.1109/TVCG.2025.3561300.
- [15] A. E. Bütün, *Semantic Normalization and Physical Realization*, Zenodo, 2026. doi:10.5281/zenodo.22947094.